SolidWorks Portfolio

Sepehr Kahvand

Master of Engineering | Certified SolidWorks Associate (CSWA) | Mechanical Designer

“Designing practical mechanical assemblies, tooling, and production-ready components with SolidWorks”

About Me

I’m a Master of Engineering graduate and Certified SolidWorks Associate with experience across part design, assemblies, surfacing, sheet metal, and motion studies. My portfolio includes practical projects such as radio-controlled helicopter, differential gearbox, car-jack mechanisms, sheet-metal kitchen sink, power supply frame, gate valve and welded structures—each focused on real-world function and manufacturability. I enjoy turning engineering concepts into clear, build-ready models and drawings that support tooling, production processes, and mechanical assembly work.

Certifications

- **CSWA – Certified SolidWorks Associate**

DASSAULT SYSTEMES

SOLIDWORKS

CERTIFICATE

Dassault Systèmes confers upon

SEPEHR KAHVAND

the certificate for

SOLIDWORKS Design Associate

(CSWA)

April 12 2026

DASSAULT SYSTEMES DESIGN SOLIDWORKS ASSOCIATE

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

C-CDQSYNGJ2T

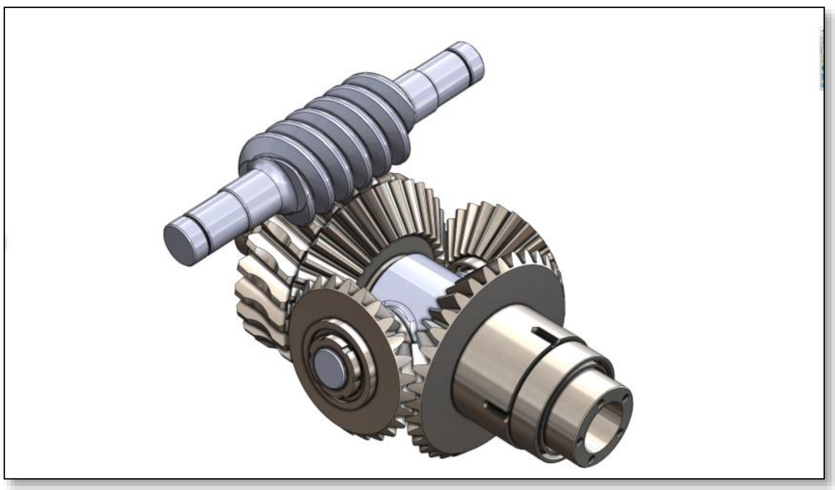
Portfolio Projects

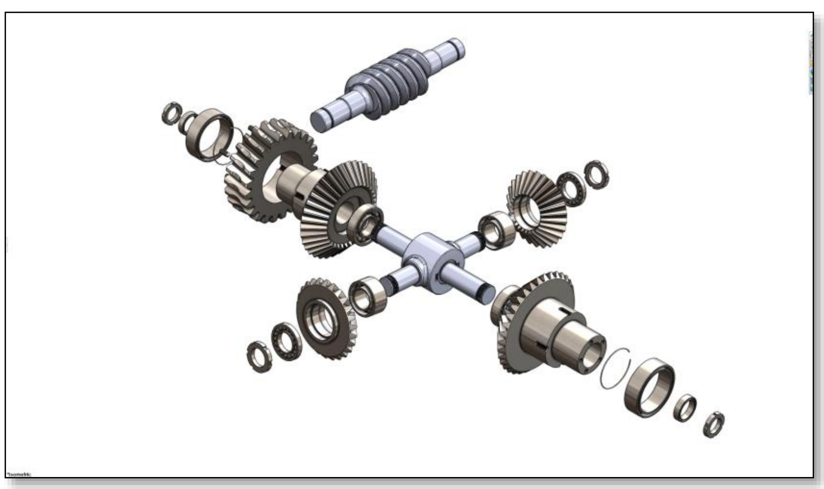
Project 1: Helicopter Rotorcraft Assembly Design

- Tools: SolidWorks part modeling, surface modeling, assemblies, mates, motion study, drafting
- Description: Developed a detailed Radio-Controlled helicopter assembly featuring a main rotor system, tail rotor drive mechanism, landing skids, transmission components, and structural frame elements. The project demonstrates advanced assembly modeling, mechanical linkage design, rotating component integration, and accurate component mating. Particular attention was given to manufacturability, assembly fit, drivetrain layout, and realistic mechanical representation of helicopter flight-control and power-transmission systems.

Project 2: Differential Gearbox

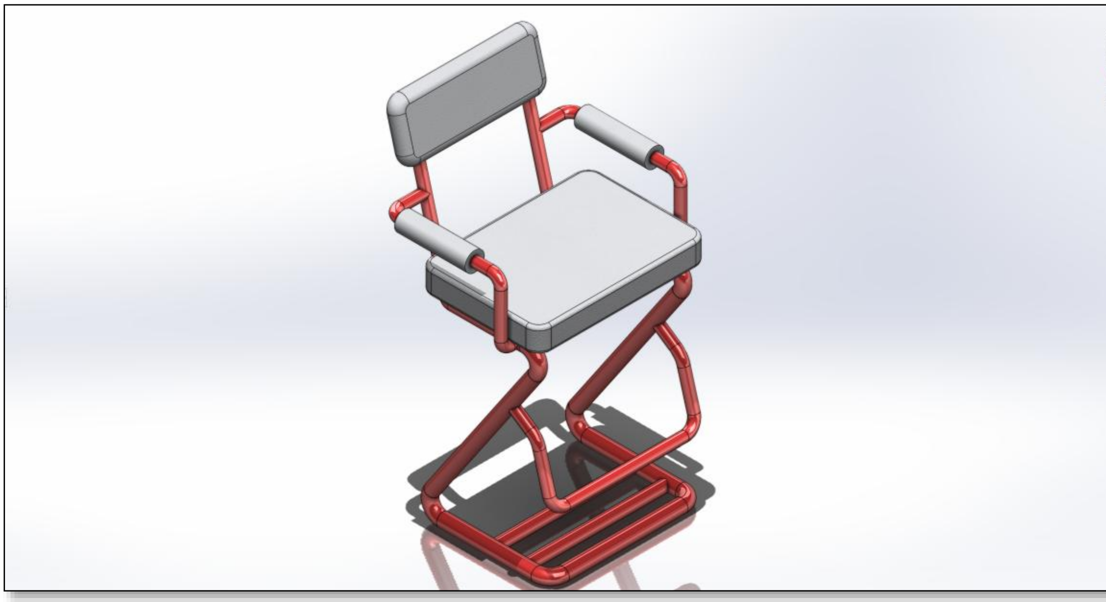
- Tools: SolidWorks part modeling, gear modeling, assemblies, mates, motion study, drafting
- Description: Developed a compact differential gearbox combining a worm-gear speed reducer with bevel-type spider gears to split torque between two output shafts. Showcases accurate gear meshing, bearing-supported shafts, and realistic multi-axis power-transmission design with strong focus on manufacturability and assembly alignment.





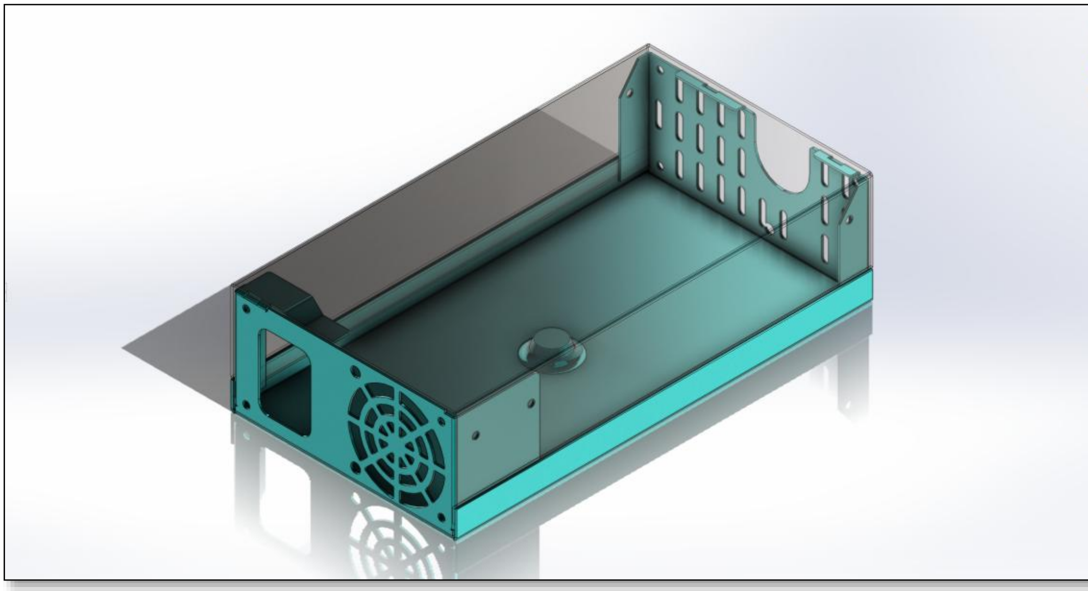
Project 3: Tubular Chair

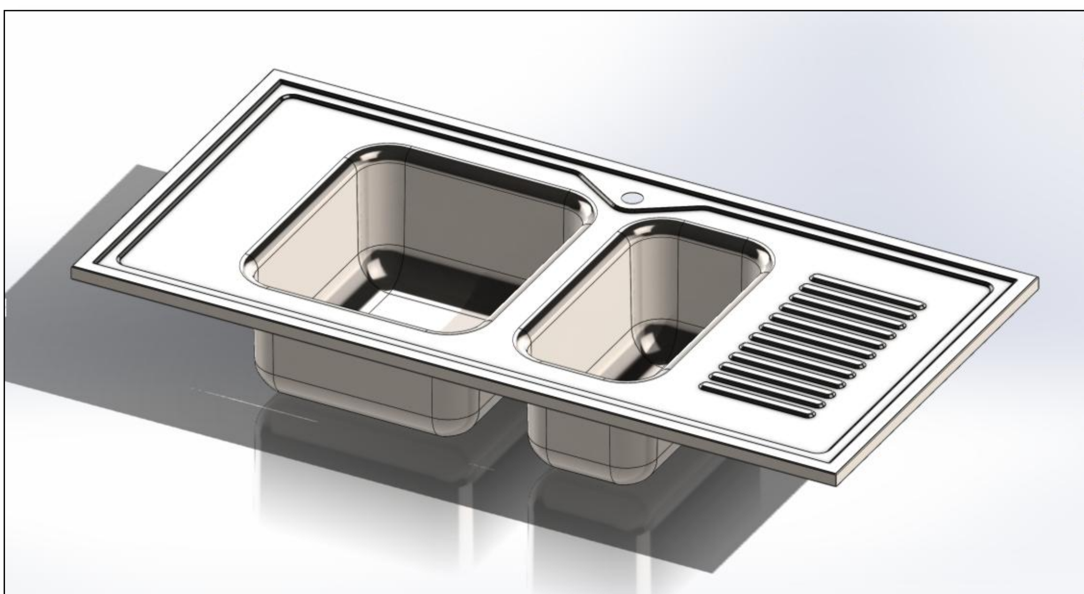
- Tools: SolidWorks Weldments, Fillet, Trim/Extend, 3D sketch
- Description: Created welded Tubular Chair using standard profiles, joints, and cut-list-driven fabrication drawings. Reflects real workstation and fixture-frame design used in automotive assembly environments.



Project 4 & 5: Power Supply Frame and Kitchen Sink

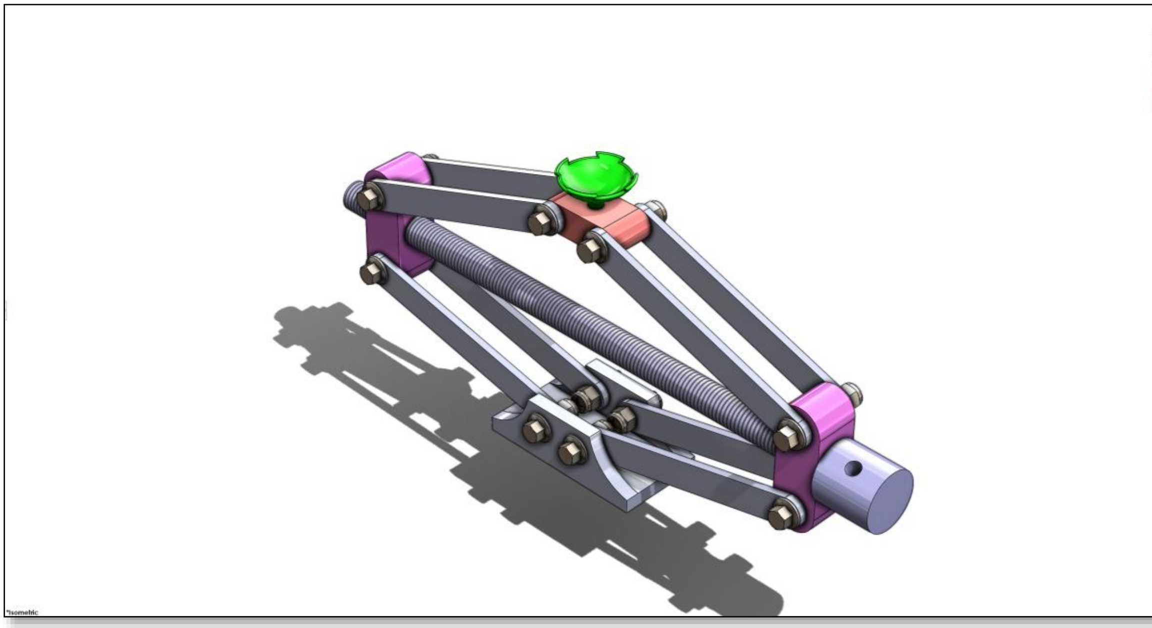
- Tools: SolidWorks Sheet Metal, Base/Edge-Flange, Forming Tools
- Description: Designed two sheet-metal components—a power-supply frame and a dual-basin sink—featuring proper bend reliefs, flanges, cutouts, and forming operations. Demonstrates strong understanding of manufacturable sheet-metal design used for enclosures, brackets, and industrial components.





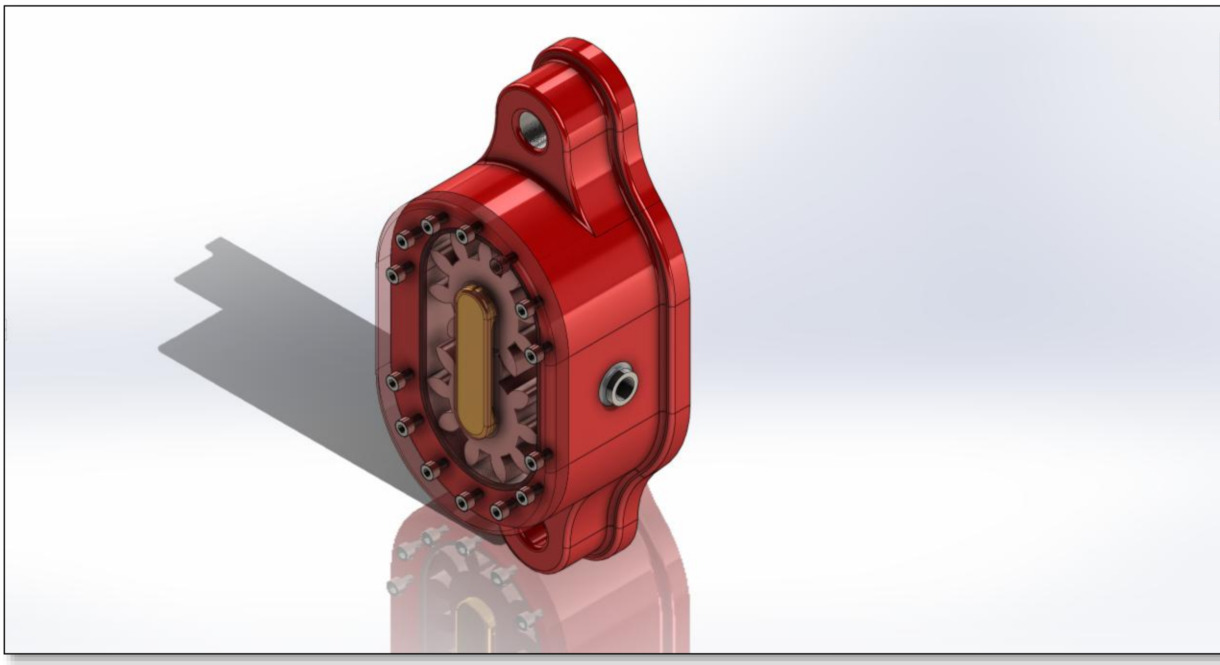
Project 6: Scissor Jack

- Tools: SolidWorks assemblies, mates, motion study
- Description: Designed a functional scissor-jack mechanism featuring linkages, pivot joints, and a central spring element. Demonstrates understanding of load paths, mechanical advantage, and motion-based mechanism design commonly used in lifting, positioning, and assist-device applications.



Project 7: Gear Pump

- Tools: SolidWorks part modeling, gear geometry, assemblies, motion
- Description: Designed a fully-modeled gear pump assembly featuring intermeshing spur gears, a sealed housing, precision-aligned shafts, and bolted end plates. Demonstrates understanding of positive-displacement pumping, torque transfer, internal clearances, and rotational motion—skills relevant to automation equipment, indexing mechanisms, and mechanical drive systems.



Project 8: Gate Valve

- Tools: SolidWorks Part Modeling, Assemblies, Section Views
- Description: Modeled a complete gate-valve assembly featuring a handwheel-driven threaded stem, internal sliding gate, flanged connections, and bolted end plates. Demonstrates understanding of fluid-control mechanisms, sealing interfaces, and precision alignment within valve bodies—skills relevant to industrial equipment, piping systems, and mechanical subsystem design.

Highlights

- 4+ years of hands-on SolidWorks experience
- Advanced skills: Part Design, Assembly,
- Intermediate Skills: Sheet Metal, Weldments, Surfaces, Motion Study, and Finite Element Analysis

Email: kahvandsepehr@gmail.com | Phone: 416-845-6397